

6_GroupByFunctions

February 11, 2025

```
[1]: import pandas as pd
```

```
[2]: df = pd.read_csv('weather_by_cities.csv')
df
```

```
[2]:
```

	day	city	temperature	windspeed	event
0	1/1/2017	new york	32	6	Rain
1	1/2/2017	new york	36	7	Sunny
2	1/3/2017	new york	28	12	Snow
3	1/4/2017	new york	33	7	Sunny
4	1/1/2017	mumbai	90	5	Sunny
5	1/2/2017	mumbai	85	12	Fog
6	1/3/2017	mumbai	87	15	Fog
7	1/4/2017	mumbai	92	5	Rain
8	1/1/2017	paris	45	20	Sunny
9	1/2/2017	paris	50	13	Cloudy
10	1/3/2017	paris	54	8	Cloudy
11	1/4/2017	paris	42	10	Cloudy

```
[4]: g = df.groupby('city')
```

```
[12]: for city,data in g:
        print('city :',city)
        print('max: ',data.temperature.max())
```

```
city : mumbai
max: 92
city : new york
max: 36
city : paris
max: 54
```

```
[14]: g['temperature'].max()
```

```
[14]: city
mumbai      92
new york    36
paris       54
```

Name: temperature, dtype: int64

```
[28]: print(df)
```

	day	city	temperature	windspeed	event
0	1/1/2017	new york	32	6	Rain
1	1/2/2017	new york	36	7	Sunny
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```
[31]: def grouper(df,idx,col):  
      if(80<=df[col].loc[idx] <= 90):  
          return '80-90'  
      elif (50 <= df[col].loc[idx] <=60):  
          return '50-60'  
      else:  
          return 'others'
```

```
[32]: f = df.groupby(lambda idx: grouper(df, idx, 'temperature'))  
      for key,d in f:  
          print('key: ', key)  
          print('data', data)
```

key: 50-60

data	day	city	temperature	windspeed	event
8	1/1/2017	paris	45	20	Sunny
9	1/2/2017	paris	50	13	Cloudy
10	1/3/2017	paris	54	8	Cloudy
11	1/4/2017	paris	42	10	Cloudy

key: 80-90

data	day	city	temperature	windspeed	event
8	1/1/2017	paris	45	20	Sunny
9	1/2/2017	paris	50	13	Cloudy
10	1/3/2017	paris	54	8	Cloudy
11	1/4/2017	paris	42	10	Cloudy

key: others

data	day	city	temperature	windspeed	event
8	1/1/2017	paris	45	20	Sunny
9	1/2/2017	paris	50	13	Cloudy
10	1/3/2017	paris	54	8	Cloudy

11 1/4/2017 paris 42 10 Cloudy

[]: